Forecasting California Industrial Petroleum CO2 Emissions

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Introduction

California publishes over five decades of industrial petroleum CO2 emissions data through the EIA, and I wanted to see how far a fairly ordinary forecasting toolkit could get on a series like this: one dominated by a long-run trend, thin on seasonality, and with real structural breaks (the 2008 recession, COVID-19) sitting in the tail end of the test period. I compare three approaches that make very different assumptions about the data - a classical SARIMA model, an L1-regularized linear model (Lasso) built on hand-engineered lag and moving-average features, and a Gradient Boosting Regressor on the same features - and I spend a good portion of this write-up on a question I did not take seriously enough the first time through: what does out-of-sample actually mean when the three models do not have access to the same information at prediction time?

The data comes from EIA's state-level emissions series (EMISS.CO2-TOTV-IC-PE-CA.A), reported annually from 1970 to 2021 and forward-filled to a monthly frequency so the models have more observations to work with. That forward-fill matters for how I read the ACF/PACF plots below - within-year values are by construction identical, so short-lag autocorrelation is partly an artifact of the resampling, not new information.

Exploratory Data Analysis

The raw series is visibly non-stationary - a long upward climb through the 1970s and 80s, a plateau, then a slower decline after the mid-2000s. An Augmented Dickey-Fuller test on the level series confirms this formally (ADF statistic -2.11, p = 0.24, well above the 0.05 threshold for rejecting a unit root), which is what pointed me toward first-differencing in the SARIMA specification below.

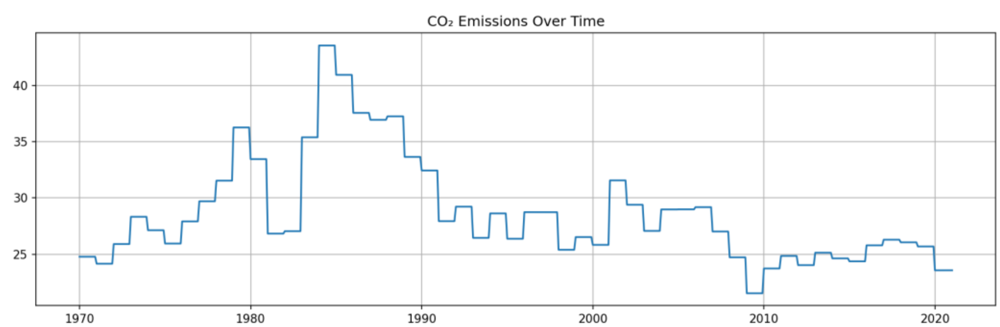

Figure 1. CA industrial petroleum CO2 emissions, monthly (forward-filled from annual data), 1970-2021.

The ACF and PACF on the original series both decay slowly rather than cutting off, which is the signature of a trending, non-stationary process rather than a case for a specific ARMA order.

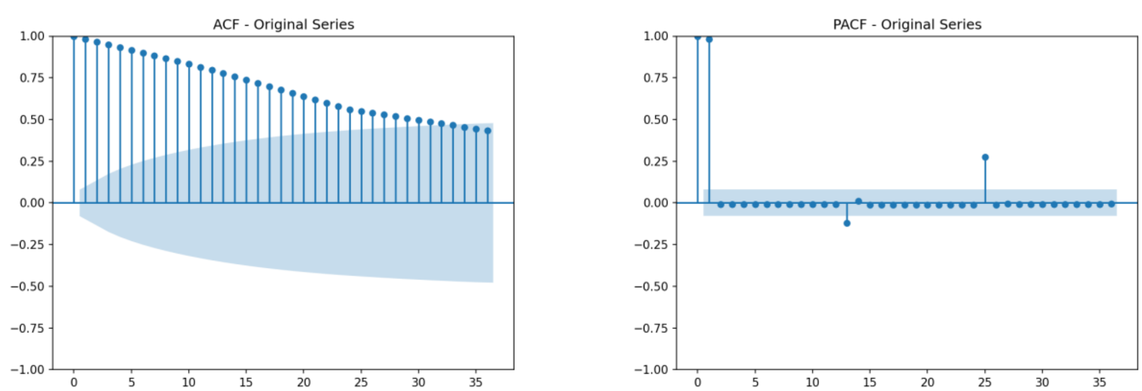

Figure 2. ACF and PACF of the undifferenced series.

A seasonal decomposition (additive, 12-month period) shows what I expected given the annual-to-monthly forward-fill: the seasonal component is essentially flat. Industrial petroleum consumption does not have a strong calendar-driven cycle the way, say, retail sales or heating demand would, so I did not carry a seasonal order into the SARIMA specification.

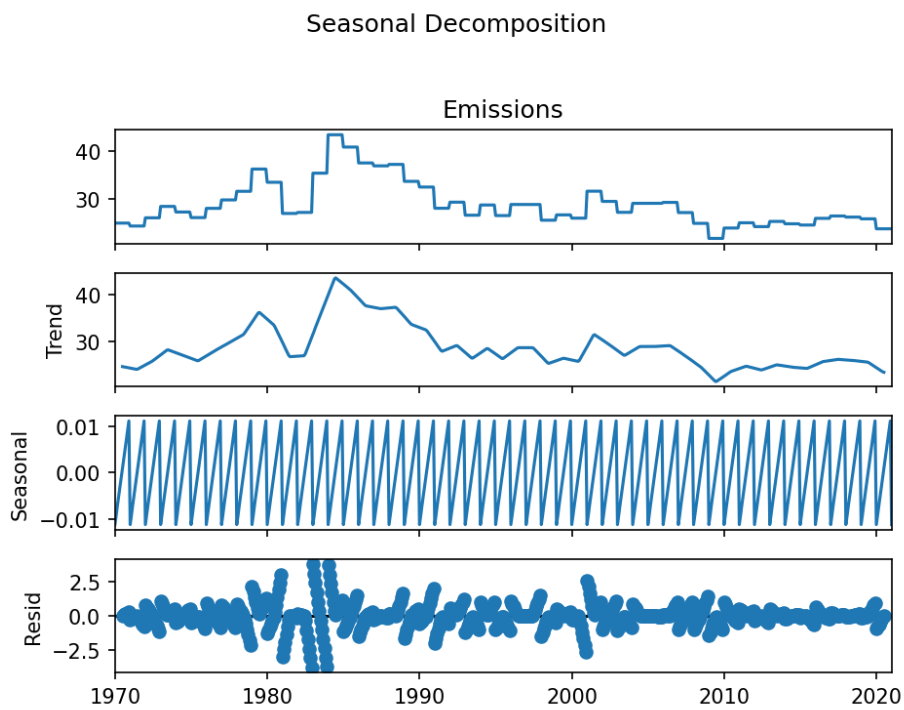

Figure 3. Additive seasonal decomposition.

Modeling Part 1: SARIMA

I fit a SARIMA(1,1,1) with no seasonal terms - first-differencing to handle the non-stationarity the ADF test flagged, an AR(1) and MA(1) term to soak up the remaining short-run autocorrelation, and no seasonal order given the flat decomposition above.

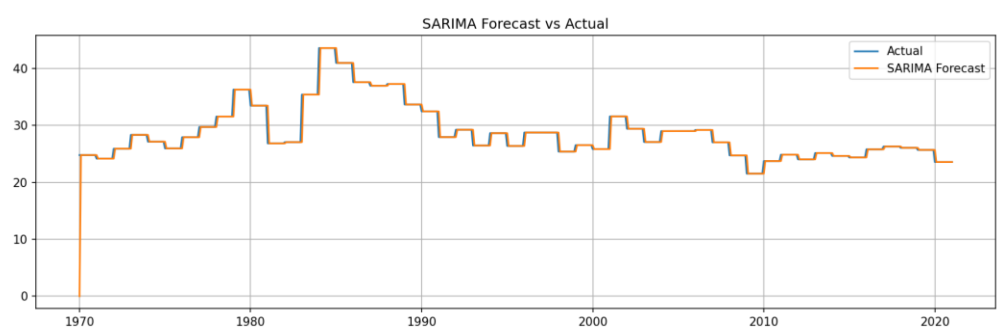

Figure 4. SARIMA fitted values against the actual series.

The residual diagnostics are about as clean as I have seen on a real dataset. The Ljung-Box test on 12 lags of the residuals returns a p-value of essentially 1.0 (0.9999), meaning there is no detectable leftover autocorrelation - the AR(1)/MA(1)/differencing structure is absorbing what needs to be absorbed. The residual histogram is close to symmetric and the residuals-vs-fitted plot does not show an obvious funnel or curve.

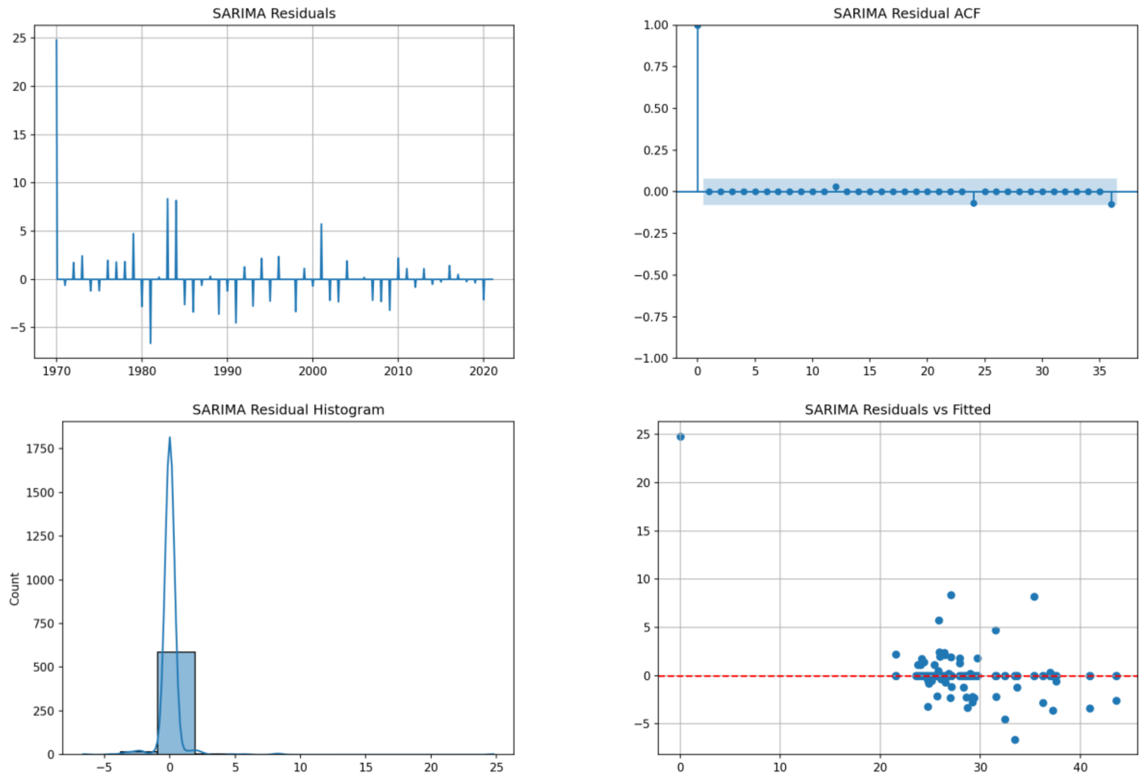

Figure 5. SARIMA residual diagnostics: residuals over time, residual ACF, residual histogram, residuals vs. fitted.

I want to flag something up front because it matters a lot for how the model comparison below should be read: the Ljung-Box test and the plots above are computed on fitted values from a model that was estimated using the entire series, 1970 through 2021. That is a legitimate way to check whether a model's assumptions are violated, but it is not the same thing as checking how well the model predicts data it has not seen. I come back to this directly in the evaluation section, because it changes the story.

Feature Engineering

For the Lasso and Gradient Boosting models, I built a small set of features meant to give a tree- or regularization-based model roughly the same information a time-series model gets for free: Lag1 and Lag24 (one month and two years back), MA3 and MA6 (rolling 3- and 6-month averages), and Month, mostly as a sanity check given the flat seasonal decomposition. Rows with missing lag or rolling values from the start of the series were dropped, which is why the modeling sample runs from 1972 rather than 1970.

Modeling Part 2: Lasso and Gradient Boosting

I split the feature-engineered series 80/20 by time - training on 1972 through early 2011 (471 months) and holding out April 2011 through January 2021 (118 months) as a genuine test set the models never see during fitting. Both Lasso (tuned via 5-fold TimeSeriesSplit cross-validation) and Gradient Boosting are trained only on the 471-month training block.

Lasso

Lasso residuals on the held-out test set are tight and centered near zero, though the Ljung-Box test ($p = 0.0047$) does detect some leftover autocorrelation - not surprising, since Lasso has no explicit mechanism for modeling serial dependence beyond whatever the lag/MA features happen to capture. A Breusch-Pagan test for heteroskedasticity comes back at $p = 0.33$, so I do not see strong evidence of non-constant residual variance.

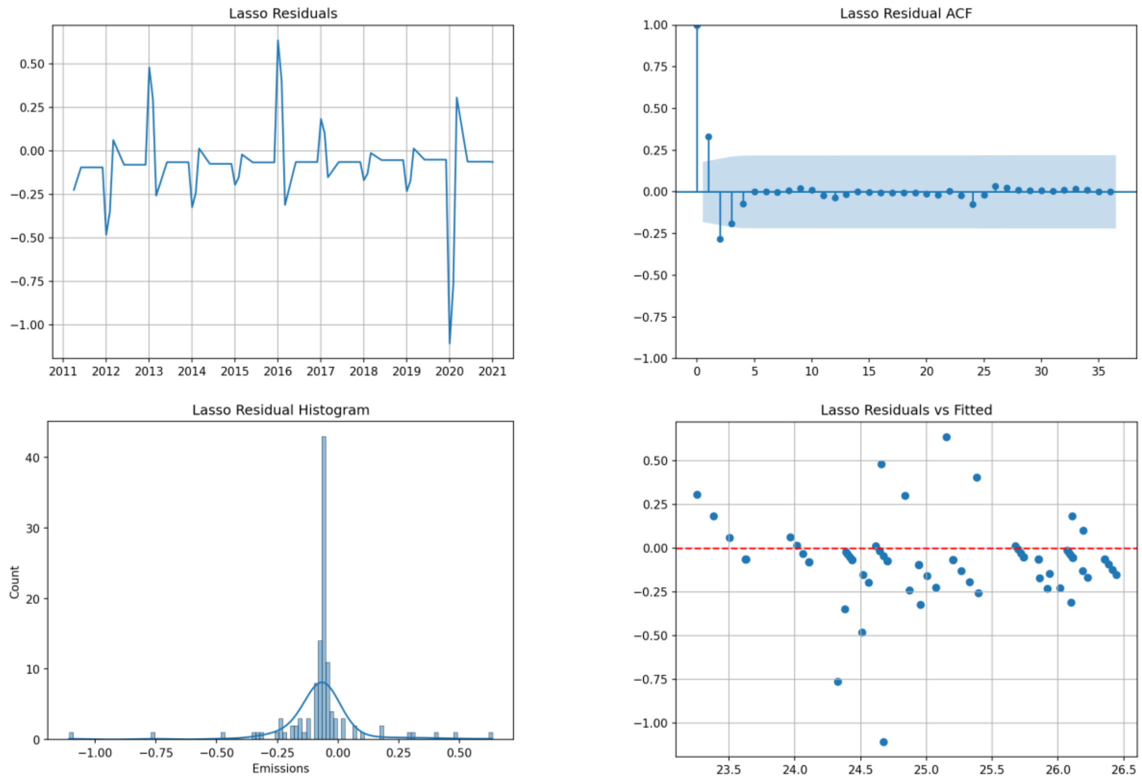

Figure 6. Lasso residual diagnostics on the held-out test set.

Gradient Boosting

Gradient Boosting is the most flexible of the three models and, per the SHAP summary below, leans almost entirely on the two moving-average features (MA3 and MA6), with Lag1 contributing little and Month confirming again that seasonality is not doing much work here. That is a useful cross-check on the feature engineering choices - the model is telling me the short-term trend features are what is carrying the signal, which is consistent with the flat decomposition from the EDA section.

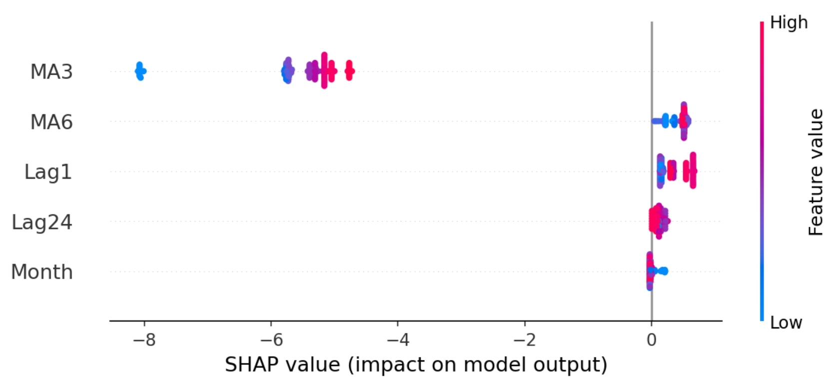

Figure 7. SHAP beeswarm plot for the Gradient Boosting model.

GBR's residual diagnostics are noticeably worse than either SARIMA or Lasso - the Ljung-Box test on GBR residuals is essentially zero (p is about 2e-36), meaning there is substantial structure left unmodeled. GBR is the most expressive model here, and that flexibility seems to come at the cost of overfitting to noise in the training window rather than generalizing cleanly.

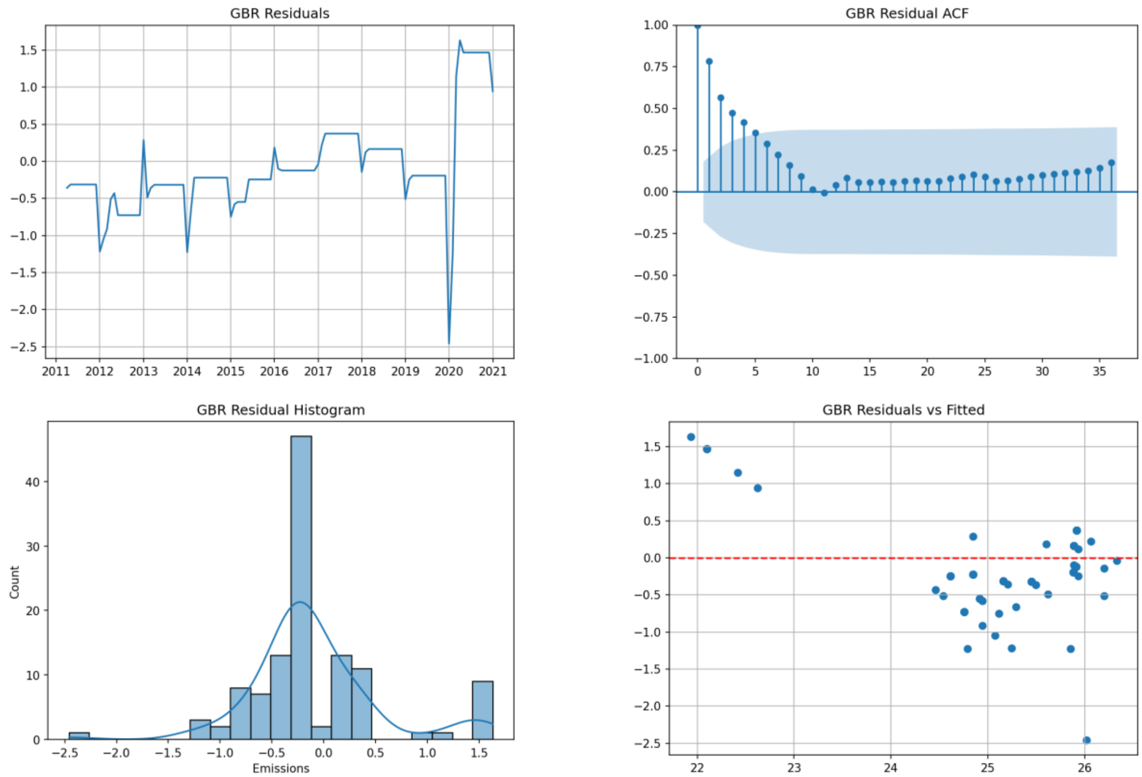

Figure 8. Gradient Boosting residual diagnostics on the held-out test set.

Model Evaluation

My first pass at comparing the three models produced the table below, and it is wrong in a way that took me longer to notice than I would like to admit.

Model	RMSE	MAE	MAPE	Evaluated on
SARIMA	0.831	0.180	0.61%	Full series, 1970-2021 (in-sample)
Lasso	0.192	0.120	0.48%	Test set only, 2011-2021 (out-of-sample)
Gradient Boosting	0.629	0.458	1.88%	Test set only, 2011-2021 (out-of-sample)

SARIMA's numbers here come from fitted values on the full series, including the years the model was estimated on. Lasso and GBR's numbers come from a genuine 118-month holdout the models never touched during training. That is not a fair fight: SARIMA is getting graded on an open-book exam while Lasso and GBR take it closed-book. It happens to still lose on RMSE even with that advantage, which is what makes it easy to miss - the conclusion that Lasso wins does not obviously look wrong. But the comparison underneath it is apples to oranges, and I do not think it is honest to leave it there.

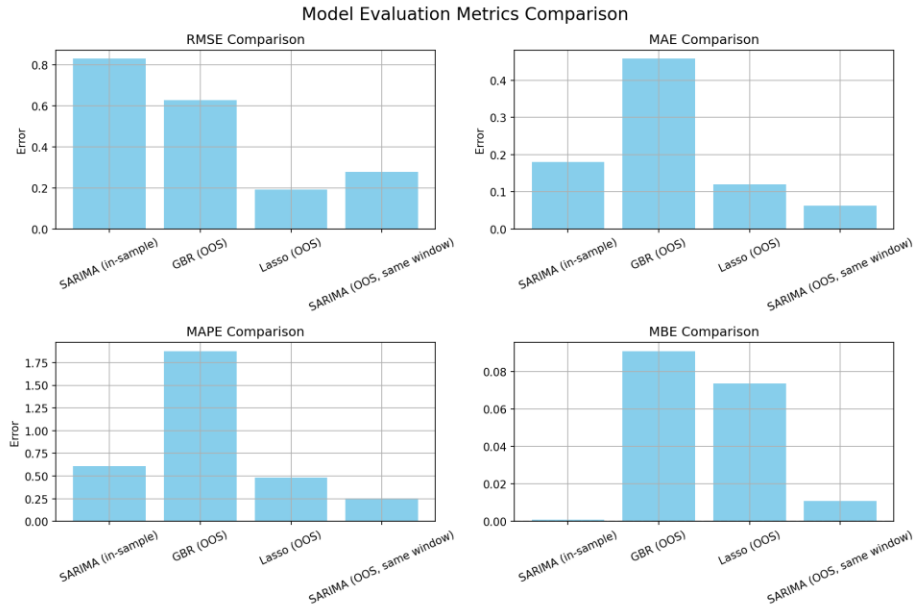

Figure 9. The original, not-quite-fair comparison across all four metrics.

A Closer Look: What Does Out-of-Sample Actually Mean Here?

Fixing this properly means asking what a fair SARIMA comparison should even look like, and it turns out there are two very different reasonable answers, depending on how much information SARIMA is allowed at prediction time.

Version 1: A genuinely blind forecast

The strictest version of out-of-sample refits SARIMA using only the training data (through March 2011) and then forecasts forward 118 months with no updating - the model never sees a single true value from the test period. Under this protocol, SARIMA's forecast converges almost immediately to a flat line around 24.85 and stays there for the full ten years, since an ARIMA(1,1,1) with no drift term reverts to something close to a random walk over long horizons. The result is a RMSE of 0.900 and a MAPE of 3.09%, worse than either Lasso or GBR, and worse than SARIMA's own misleading in-sample number.

Version 2: A one-step-ahead forecast with the same information Lasso gets

But that is arguably not a fair comparison either, in the other direction: Lasso and GBR are not doing blind 118-month-ahead forecasting. Their Lag1, Lag24, MA3, and MA6 features are computed from the true, actual values at each point in the test period, so at every test month they effectively get to see last month's real emissions level before predicting this month. A SARIMA model given the equivalent privilege - parameters estimated from training data only, but then generating one-step-ahead predictions using the true prior values in the test window, the same way Lasso's Lag1 feature does - tells a different story: RMSE 0.279, MAE 0.062, MAPE 0.25%. On MAE and MAPE, that beats Lasso outright. On RMSE, Lasso still edges it out, 0.192 to 0.279, likely because Lasso has fewer large misses in percentage terms even if its typical error is bigger.

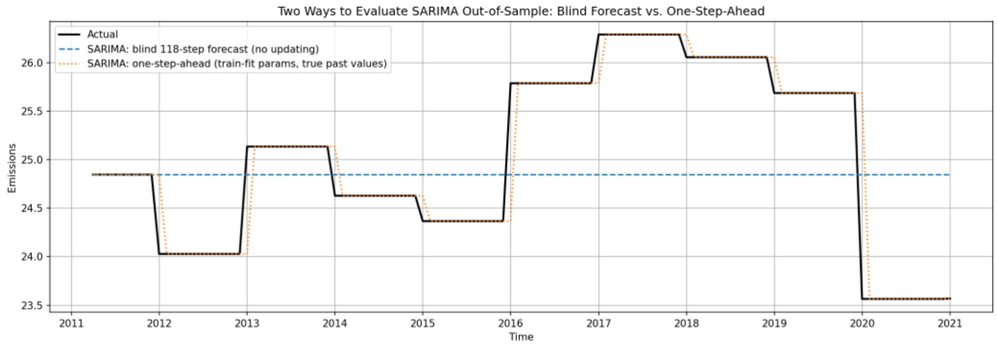

Figure 10. Two very different SARIMA forecasts over the same test window: a blind 118-month-ahead extrapolation versus a one-step-ahead forecast using true prior values.

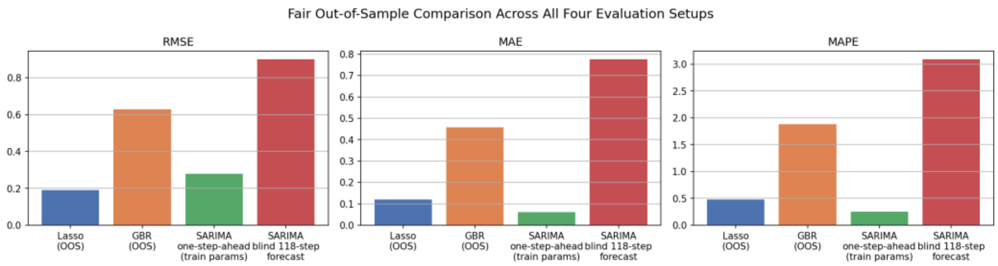

Figure 11. RMSE, MAE, and MAPE across all four evaluation setups, side by side.

I think this is the more useful finding than either of my earlier passes. Which model is best is not a fixed answer here - it depends on what the forecasting task actually is. If the real use case is a long-range extrapolation with no opportunity to update on new data as it arrives, Lasso and GBR's engineered features give them a real, durable edge over SARIMA, which just flattens out. If the use case looks more like rolling, one-step-ahead forecasting, updating each month as new data comes in, SARIMA is competitive with, and by some metrics better than, the machine learning models, largely because its differencing term is doing something similar to what Lasso's Lag1 feature is doing by hand. The original conclusion that Lasso wins is directionally fine for the blind-forecast case and actively misleading for the rolling-forecast case, and the write-up should not have collapsed those into one number.

Post-Hoc Analysis

Looking at where the models actually diverge from the truth over time, using each model's respective test-period predictions, all three track the broad shape of the series reasonably well through the 2000s, with more daylight opening up after 2010, particularly around the 2020 COVID-related emissions drop, which none of the models were given any exogenous signal to anticipate.

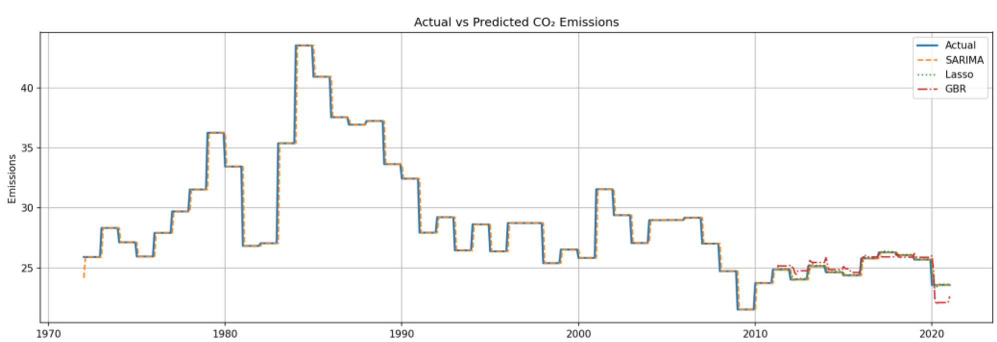

Figure 12. Actual emissions against each model's predictions.

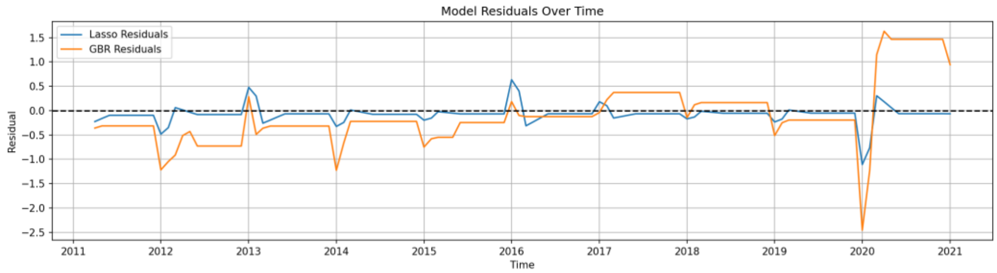

Figure 13. Lasso and GBR residuals over the test window.

The cumulative forecast error tells a similar story with more texture: Lasso drifts down gradually and persistently, which reads as a mild, consistent negative bias from L1 shrinkage rather than any one bad stretch, while GBR's cumulative error swings much harder - a steep decline through about 2016, a partial correction, then a sharp break around the 2020 disruption. GBR is more reactive to real shocks when they happen, but that same flexibility makes it noisier and harder to trust between shocks.

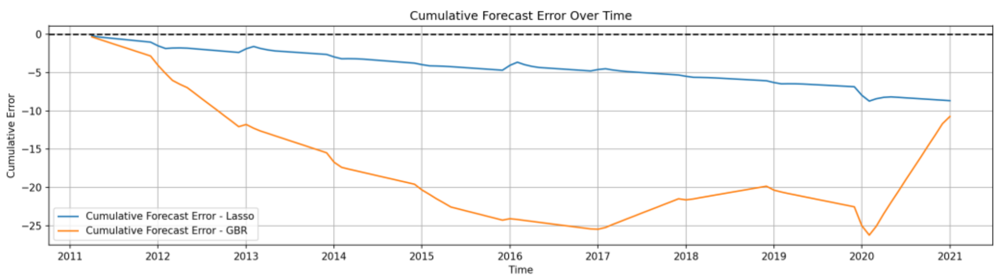

Figure 14. Cumulative forecast error, Lasso vs. GBR.

What the Best Model Is Actually Using

Lasso's cross-validated regularization parameter (alpha = 0.0047) is small enough that it barely shrinks anything - all four engineered features survive with non-zero coefficients, led by MA3 (7.93), followed by MA6 (-1.62) and Lag1 (-1.62) at roughly equal and opposite magnitude, with Lag24 contributing only a small amount (-0.04). The dominant role of the short moving average and the near-cancellation between MA6 and Lag1 both point toward the same conclusion as the GBR SHAP plot: almost all of the predictive power in this series comes from very recent momentum, not from anything resembling a yearly cycle.

Conclusion

Three models, three different pictures of the same series. SARIMA has the cleanest residual diagnostics of the three by a wide margin, essentially no leftover autocorrelation, but that cleanliness is partly a function of being checked against data it was fit on. Lasso is the most reliable model for a genuine long-horizon forecast, and its regularization path also does useful work as a feature-selection tool: four simple engineered features get you most of the way there. Gradient Boosting is the most reactive to real structural breaks like 2020 but pays for that flexibility with the weakest residual diagnostics of the group.

The bigger lesson for me was less about which model wins and more about how easy it is to build an evaluation that quietly favors one model over another without meaning to. Catching the in-sample/out-of-sample mismatch was step one; realizing that fixing it naively, with a blind long-horizon forecast, just traded one unfair comparison for a different one was the part I almost missed. I would rather report both fair comparisons and let the trade-off be visible than collapse it into a single best-model claim that only holds under one specific forecasting protocol.